



TCC805x MCU BSP

API Specification

for General Purpose Serial Bus Controller

Rev. 1.01 [A]

2021-01-22

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1 INTRODUCTION

This document describes the general purpose serial bus controller device driver (hereinafter referred to as GPSB driver) of TCC805x MCU BSP. For more detailed information of the general purpose serial bus controller in the TCC805x MCU Sub-system, refer to Chapter 11 General Purpose Serial Bus (GPSB) Controller of the *“PART 11-MICOM SUB-SYSTEM”* in the *“TCC805x Full Specification”*. [1]

2 TERMS AND ABBREVIATIONS

Table 2.1 Terms and Abbreviation

GPSB	General Purpose Serial Bus
BPW	Bit Per Width
SCK	GPSB(SPI) Clock
SDO	GPSB(SPI) Output (Master channel output, Slave channel is <u>not</u> supported.)
SDI	GPSB(SPI) Input (Master channel input, Slave channel is <u>not</u> supported.)
CS	Chip Select PIN of GPSB
SDM	Safe Display Manager
RVC	Rear Video Camera

3 SOURCE FILES

3.1 Location of Source Files

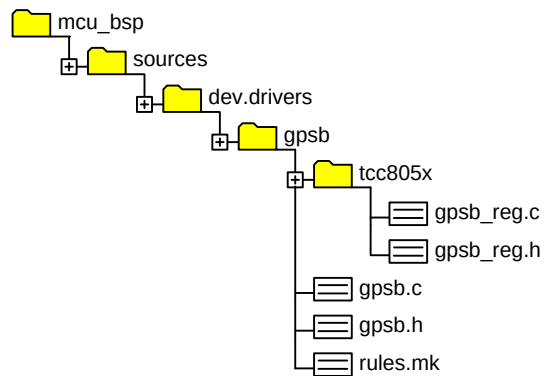


Figure 3.1 Location of the source files

3.2 Simple Description of Source Files

- `gpsb_reg.c`
 - GPSB specific source code by chipset.
- `gpsb_reg.h`
 - GPSB specific header by chipset.
- `gpsb.c`
 - GPSB common source code.
- `gpsb.h`
 - GPSB common header including API & DEFINE for user
- `rules.mk`
 - Rules used for build

4 CONSTANTS

4.1 GPSB Base Address

Constants values for base register address for each GPSB channel and GPSB configuration. They are used to change register value for the GPSB operation.

Declaration

```
#define TCC_MICOM_GPSB_BASE
#define GPSB0_BASE      (TCC_MICOM_GPSB_BASE + 0x00000)
#define GPSB1_BASE      (TCC_MICOM_GPSB_BASE + 0x10000)
#define GPSB2_BASE      (TCC_MICOM_GPSB_BASE + 0x20000)
#define GPSB3_BASE      (TCC_MICOM_GPSB_BASE + 0x30000)
#define GPSB4_BASE      (TCC_MICOM_GPSB_BASE + 0x40000)
#define GPSB5_BASE      (TCC_MICOM_GPSB_BASE + 0x50000)
#define GPSB_PCFG_BASE   (TCC_MICOM_GPSB_BASE + 0x60000)
#define GPSB_IO_MONITOR_BASE (TCC_MICOM_GPSB_BASE + 0x70000)
```

Description

Value	Description
TCC_MICOM_GPSB_BASE	GPSB controller register base address: 0x1B10000UL
GPSB0_BASE	GPSB controller channel 0 base address: 0x1B10000UL
GPSB1_BASE	GPSB controller channel 1 base address: 0x1B11000UL
GPSB2_BASE	GPSB controller channel 2 base address: 0x1B12000UL
GPSB3_BASE	GPSB controller channel 3 base address: 0x1B13000UL
GPSB4_BASE	GPSB controller channel 4 base address: 0x1B14000UL
GPSB5_BASE	GPSB controller channel 5 base address: 0x1B15000UL
GPSB_PCFG_BASE	GPSB controller port configuration base address: 0x1B16000UL
GPSB_IO_MONITOR_BASE	GPSB controller IO monitor base address: 0x1B17000UL

Remark

None

4.2 GPSB Channel

Constants values for index of each channel.

Declaration

```
#define GPSB_CH_0    (0UL)
#define GPSB_CH_1    (1UL)
#define GPSB_CH_2    (2UL)
#define GPSB_CH_3    (3UL)
#define GPSB_CH_4    (4UL)
```

Description

Value	Description
<i>GPSB_CH_0</i>	<i>GPSB Channel 0 index value</i>
<i>GPSB_CH_1</i>	<i>GPSB Channel 1 index value</i>
<i>GPSB_CH_2</i>	<i>GPSB Channel 2 index value</i>
<i>GPSB_CH_3</i>	<i>GPSB Channel 3 index value. (This channel is reserved for SDM. If you want to use this channel for normal purpose, change SDM channel value to 0x6 or 0x7 via calling API GPSB_SetSdmChaanel.)</i>
<i>GPSB_CH_4</i>	<i>GPSB Channel 4 index value. (This channel is reserved for SRVC. If you want to use this channel for normal purpose, change SDM channel value to 0x6 or 0x7 via calling API GPSB_SetRvcChaanel.)</i>

Remark

Refer to *SDM_RVC_CTRL_SEL* in "*TCC805x Full Specification*". [1]

4.3 GPSB Mode

Constants values for the GPSB transfer mode as parameter of [GPSB_Xfer](#) function.

Declaration

```
#define GPSB_XFER_MODE_WITHOUT_INTERRUPT    (uint32)(0x01UL)
#define GPSB_XFER_MODE_WITH_INTERRUPT      (uint32)(0x02UL)
#define GPSB_XFER_MODE_DMA                 (uint32)(0x10UL)
#define GPSB_XFER_MODE_PIO                 (uint32)(0x20UL)
#define GPSB_XFER_MODE_WITHOUT_CTF        (uint32)(0x40UL)
```

Description

Value	Description
<i><code>GPSB_XFER_MODE_WITHOUT_INTERRUPT</code></i>	<i>Non-interrupt transfer mode</i>
<i><code>GPSB_XFER_MODE_WITH_INTERRUPT</code></i>	<i>Interrupt transfer mode</i>
<i><code>GPSB_XFER_MODE_DMA</code></i>	<i>DMA transfer mode</i>
<i><code>GPSB_XFER_MODE_PIO</code></i>	<i>PIO (FIFO) transfer mode</i>
<i><code>GPSB_XFER_MODE_WITHOUT_CTF</code></i>	<i>Continuous transfer mode (Keep CS signal activate state)</i>

Remark

None

4.4 GPSB Event

Constants values for the GPSB error event. When the GPSB error interrupt occurs, the GPSB driver notifies a user of the errors via callback function.

Declaration

```
#define GPSB_EVENT_COMPLETE (uint32)(0x0000UL)
#define GPSB_EVENT_ERR_ROR (uint32)(0x0100UL)
#define GPSB_EVENT_ERR_WUR (uint32)(0x0200UL)
#define GPSB_EVENT_ERR_RUR (uint32)(0x0400UL)
#define GPSB_EVENT_ERR_WOR (uint32)(0x0800UL)
```

Description

Value	Description
<i>GPSB_EVENT_COMPLETE</i>	<i>Transfer complete</i>
<i>GPSB_EVENT_ERR_ROR</i>	<i>Read FIFO over-run error</i>
<i>GPSB_EVENT_ERR_WUR</i>	<i>Write FIFO under-run error</i>
<i>GPSB_EVENT_ERR_RUR</i>	<i>Read FIFO under-run error</i>
<i>GPSB_EVENT_ERR_WOR</i>	<i>Write FIFO over-run error</i>

Remark

None

4.5 GPSB State

Constants values for the GPSB transfer state management.

Declaration

```
#define GPSB_XFER_STATE_DISABLED    (uint8)(0x00UL)
#define GPSB_XFER_STATE_IDLE       (uint8)(0x01UL)
#define GPSB_XFER_STATE_RUNNING    (uint8)(0x02UL)
```

Description

Value	Description
<i>GPSB_XFER_STATE_DISABLED</i>	<i>GPSB is disabled.</i>
<i>GPSB_XFER_STATE_IDLE</i>	<i>GPSB is in idle state.</i>
<i>GPSB_XFER_STATE_RUNNING</i>	<i>GPSB is in running state. Other transfer requests are blocked in this state.</i>

Remark

None

5 STRUCTURES

5.1 GPSBDev_t

Structure to store operating information of each GPSB channel. GPSBDev_t is declared in *gpsb_reg.c* and has a default initial value. If you input parameter values when calling API such as *GPSB_Open*, the parameter values are automatically stored in this struct member variable.

Declaration

```
typedef struct _GPSBDev_  
{  
    Uint32          dChannel;  
    uint32          dBase;  
    uint32          dCfgAddr;  
    uint32          dPeriName;  
    uint32          dIobusName;  
    uint32          dIsSlave;  
    uint32          dSdo;  
    uint32          dSdi;  
    uint32          dSclk;  
    uint32          dPort;  
    uint32          dIntrPriority;  
    uint32          dSupportDma;  
    GPSBDmaBuff_t   dTxDma;  
    GPSBDmaBuff_t   dRxDma;  
    const void *     dAsyncTxBuf;  
    void *           dAsyncRxBuf;  
    uint32          dAsyncTxDataSize;  
    uint32          dAsyncRxDataSize;  
    uint8           dState;  
    GPSBXferComplite_t dComplete;  
} GPSBDev_t;
```

Description

Value	Description
<i>dChannel</i>	<i>GPSB channel number</i>
<i>dBase</i>	<i>GPSB channel base address</i>
<i>dCfgAddr</i>	<i>GPSB port configuration address</i>
<i>dPeriName</i>	<i>Peri clock index</i>
<i>dIobusName</i>	<i>IOBUS index</i>
<i>dIsSlave</i>	<i>GPSB mode (Master or Slave)</i>
<i>dSdo</i>	<i>SDO(MOSI) GPIO index info.</i>
<i>dSdi</i>	<i>SDI(MISO) GPIO index info.</i>
<i>dSclk</i>	<i>SCLK GPIO index info.</i>
<i>dPort</i>	<i>Matched port number</i>
<i>dIntrPriority</i>	<i>Interrupt priority</i>
<i>dSupportDma</i>	<i>Support info. Default: FALSE</i>
<i>dTxDma</i>	<i>Tx DMA Buffer info.</i>
<i>dRxDma</i>	<i>Rx DMA Buffer info.</i>
<i>dAsyncTxBuf</i>	<i>Tx buffer address for FIFO async transfer</i>
<i>dAsyncRxBuf</i>	<i>Rx buffer address for FIFO async transfer</i>
<i>dAsyncTxDataSize</i>	<i>Async FIFO transfer Tx data size</i>
<i>dAsyncRxDataSize</i>	<i>Async FIFO transfer Rx data size</i>
<i>dState</i>	<i>State of GPSB</i>
<i>dComplete</i>	<i>Notify the completion of non-blocking transfer.</i>

Remark

None

5.2 GPSBXferComplete_t

Structure to register callback function. The value of *xcCallback* parameters is filled when calling *GPSB_Open*.

Declaration

```
typedef struct _GPSBXferComplete_  
{  
    GPSBCallback    xcCallback;  
    void *          xcArg;  
} GPSBXferComplite_t;
```

Description

Value	Description
<i>xcCallback</i>	Callback function pointer. Refer to GPSBCallback .
<i>xcArg</i>	Etc.

Remark

None

6 CALLBACK FUNCTIONS

6.1 GPSBCallback

When async transfer is completed or an error interrupt occurs, the GPSB driver notifies a user of the event information via callback function.

The supported events are as follows:

- Transmit done.
- Error interrupts.

Declaration

```
typedef void (*GPSBCallback)
(
    uint32      uiCh,
    int32       iEvent,
    void *      pArg
);
```

Parameters

Value	Description
<i>uiCh</i>	Value of channel to be controlled
<i>iEvent</i>	Value of Event information. Refer to GPSB Event .
<i>pArg</i>	Parameter from user

Return

None

Remark

None

7 APIs

7.1 GPSB_Init

Registers the GPSB interrupt handler functions and enables the GPSB interrupts. Normally, interrupts occur when using FIFO or DMA async transfer.

Supported interrupts: GPSB 0~4, GPSB DMA 0~4

Declaration

```
void GPSB_Init  
(  
    void  
);
```

Parameters

None

Return

None

Remark

None

7.2 GPSB_Deinit

Disables all the GPSB interrupts.

Declaration

```
void GPSB_Deinit  
(  
    void  
);
```

Parameters

None

Return

None

Remark

None

7.3 GPSB_Reset

Resets the GPSB channel in IOBUS. After reset, the GPSB channel register is set to default value.

Declaration

```
SALRetCode_t GPSB_Reset  
(  
    Uint32    uiCh  
);
```

Parameters

Value	Description
uiCh	Value of channel to be controlled

Return

Value	Description
SAL_RET_SUCCESS	Success
SAL_RET_FAILED	Failure

Remark

None

7.4 GPSB_Open

Initializes the GPSB channel. Input valid parameter to operate the GPSB channel. SCK/SDO/SDI values depend on HW construction. (If use internal path such as loopback mode, can set SDO/SDI/SCK to 0.) If want to get notification from the GPSB driver, register callback function.

Declaration

```
SALRetCode_t GPSB_Open
(
    uint32      uiCh,
    uint32      uiSdo,
    uint32      uiSdi,
    uint32      uiSclk,
    uint32      uiDmaAddrTx,
    uint32      uiDmaAddrRx,
    uint32      uiDmaBufSize,
    GPSBCallback fbCallback,
    void *      pArg
);
```

Parameters

Value	Description
<i>uiCh</i>	<i>Value of channel to be controlled</i>
<i>uiSdo</i>	<i>SDO GPIO index</i>
<i>uiSdi</i>	<i>SDI GPIO index</i>
<i>uiSclk</i>	<i>SCLK GPIO index</i>
<i>uiDmaAddrTx</i>	<i>DMA TX buffer physical address</i>
<i>uiDmaAddrRx</i>	<i>DMA RX buffer physical address</i>
<i>uiDmaBufSize</i>	<i>DMA buffer size</i>
<i>fbCallback</i>	<i>Callback function pointer. Refer to the GPSBCallback.</i>
<i>pArg</i>	<i>Etc.</i>

Return

Value	Description
<i>SAL_RET_SUCCESS</i>	<i>Success</i>
<i>SAL_RET_FAILED</i>	<i>Failure</i>

Remark

None

7.5 GPSB_Close

De-initializes GPSB channel.

- Stop DMA.
- Disable channel interrupt.
- Clear register value to disable GPSB channel.

Declaration

```
SALRetCode_t GPSB_Close  
(  
    uint32    uiCh  
);
```

Parameters

Value	Description
uiCh	Value of channel to be controlled

Return

Value	Description
SAL_RET_SUCCESS	Success
SAL_RET_FAILED	Failure

Remark

None

7.6 GPSB_SetSpeed

Sets GPSB operating clock speed. The maximum operating clock speed is limited by HW construction and signal strength.

Declaration

```
SALRetCode_t GPSB_SetSpeed  
(  
    uint32    uiCh,  
    uint32    uiSpeedInHz  
);
```

Parameters

Value	Description
<i>uiCh</i>	<i>Value of channel to be controlled</i>
<i>uiSpeedInHz</i>	<i>SCLK speed</i>

Return

Value	Description
<i>SAL_RET_SUCCESS</i>	<i>Success</i>
<i>SAL_RET_FAILED</i>	<i>Failure</i>

Remark

None

7.7 GPSB_SetBpw

Sets the BPW value. Valid BPW value is 8, 16, or 32.

Declaration

```
SALRetCode_t GPSB_SetBpw  
(  
    uint32    uiCh,  
    uint32    uiBpw  
);
```

Parameters

Value	Description
<i>uiCh</i>	<i>Value of channel to be controlled</i>
<i>uiBpw</i>	<i>Bit per word</i>

Return

Value	Description
<i>SAL_RET_SUCCESS</i>	<i>Success</i>
<i>SAL_RET_FAILED</i>	<i>Failure</i>

Remark

None

7.8 GPSB_SetMode

Sets the GPSB operating mode. Polarity and phase of the GPSB signal change according to the mode value. Refer to GPSB MODE register & Timing Diagram in “TCC805x Full Specification”.

Declaration

```
void GPSB_SetMode
(
    uint32    uiCh,
    uint32    uiMode
);
```

Parameters

Value	Description
uiCh	Value of channel to be controlled
uiMode	Operating Mode. (e.g. GPSB_CPOL, GPSB_CPHA, GPSB_LOOP, etc.)

Return

None

Remark

None

7.9 GPSB_CsInit

Initializes CS GPIO to default config.

Declaration

```
SALRetCode_t GPSB_CsInit  
(  
    uint32    uiCh,  
    uint32    uiCs,  
    uint32    uiCsHigh  
);
```

Parameters

Value	Description
<i>uiCh</i>	<i>Value of channel to be controlled</i>
<i>uiCs</i>	<i>CS GPIO index</i>
<i>uiCsHigh</i>	<i>Default high or low</i>

Return

Value	Description
<i>SAL_RET_SUCCESS</i>	<i>Success</i>
<i>SAL_RET_FAILED</i>	<i>Failure</i>

Remark

None

7.10 GPSB_CsAlloc

Configures the CS GPIO.

Declaration

```
SALRetCode_t GPSB_CsAlloc  
(  
    uint32    uiCh,  
    uint32    uiCs  
);
```

Parameters

Value	Description
<i>uiCh</i>	<i>Value of channel to be controlled</i>
<i>uiCs</i>	<i>CS GPIO index</i>

Return

Value	Description
<i>SAL_RET_SUCCESS</i>	<i>Success</i>
<i>SAL_RET_FAILED</i>	<i>Failure</i>

Remark

None

7.11 GPSB_CsActivate

Activates the CS GPIO.

Declaration

```
SALRetCode_t GPSB_CsActivate  
(  
    uint32    uiCh,  
    uint32    uiCs,  
    uint32    uiCsHigh  
);
```

Parameters

Value	Description
<i>uiCh</i>	<i>Value of channel to be controlled</i>
<i>uiCs</i>	<i>CS GPIO index</i>
<i>uiCsHigh</i>	<i>Default High or Low</i>

Return

Value	Description
<i>SAL_RET_SUCCESS</i>	<i>Success</i>
<i>SAL_RET_FAILED</i>	<i>Failure</i>

Remark

None

7.12 GPSB_CsDeactivate

Deactivates the CS GPIO.

Declaration

```
SALRetCode_t GPSB_CsDeactivate  
(  
    uint32    uiCh,  
    uint32    uiCs,  
    uint32    uiCsHigh  
);
```

Parameters

Value	Description
<i>uiCh</i>	<i>Value of channel to be controlled</i>
<i>uiCs</i>	<i>CS GPIO index</i>
<i>uiCsHigh</i>	<i>Default High or Low</i>

Return

Value	Description
<i>SAL_RET_SUCCESS</i>	<i>Success</i>
<i>SAL_RET_FAILED</i>	<i>Failure</i>

Remark

None

7.13 GPSB_Xfer

Transmits data. This function can be called from application. Before calling this function, the GPSB channel must set initializing and configuration (speed, BPW, etc.).

Declaration

```
SALRetCode_t GPSB_Xfer
(
    uint32      uiCh,
    const void * pTx,
    void *      pRx,
    uint32      uiDataSize,
    uint32      uiXferMode
);
```

Parameters

Value	Description
<i>uiCh</i>	<i>Value of channel to be controlled</i>
<i>pTx</i>	<i>TX data buffer</i>
<i>pRx</i>	<i>RX data buffer</i>
<i>uiDataSize</i>	<i>Write/Read data size</i>
<i>uiXferMode</i>	<i>Transfer mode (DMA/PIO, Interrupt/Non-Interrupt)</i>

Return

Value	Description
<i>SAL_RET_SUCCESS</i>	<i>Success</i>
<i>SAL_RET_FAILED</i>	<i>Failure</i>

Remark

None

7.14 GPSB_XferAbort

Stops transmission.

Declaration

```
void GPSB_XferAbort  
(  
    uint32    uiCh  
);
```

Parameters

Value	Description
uiCh	Value of channel to be controlled

Return

Value	Description
SAL_RET_SUCCESS	Success
SAL_RET_FAILED	Failure

Remark

None

7.15 GPSB_SetSdmChannel

Sets SDM channel. The SDM default channel is *GPSB_CH_3*.

Declaration

```
SALRetCode_t GPSB_SetSdmChannel  
(  
    uint32    uiCh  
);
```

Parameters

Value	Description
<i>uiCh</i>	<i>Value of channel to be controlled</i>

Return

Value	Description
<i>SAL_RET_SUCCESS</i>	<i>Success</i>
<i>SAL_RET_FAILED</i>	<i>Failure</i>

Remark

None

7.16 GPSB_SetRvcChannel

Sets RVC channel. The default channel of RVC is *GPSB_CH_4*.

Declaration

```
SALRetCode_t GPSB_SetRvcChannel  
(  
    uint32    uiCh  
);
```

Parameters

Value	Description
<i>uiCh</i>	<i>Value of channel to be controlled</i>

Return

Value	Description
<i>SAL_RET_SUCCESS</i>	<i>Success</i>
<i>SAL_RET_FAILED</i>	<i>Failure</i>

Remark

None

8 HOW TO USE GPSB DRIVER

GPSB driver requires channel initializing & de-initializing step before and after the data transfer.

[Initialize]

Normally, you can set up channels via calling ***GPSB_Open*** with valid parameters. If external path is used, not internal path such as loopback, you should set SCK, SDI, SDO, and CS GPIO. Operating values such as speed, Bit Per Word (BPW), and mode can be set after completing ***GPSB_Open*** without error.

[Data transfer]

After initializing, data can be transferred via calling ***GPSB_Xfer***. ***GPSB_Xfer*** requires valid buffer and xfer mode information. Before and after calling ***GPSB_Xfer***, if CTF mode is not used, you should set CS GPIO to control xfer start and stop. Then call ***GPSB_CsActivate*** & ***GPSB_CsDeactivate*** to control the CS GPIO.

[De-initialize]

After the use of GPSB channel is completed, close GPSB channel via calling ***GPSB_Close*** to prevent useless resource and to avoid errors caused by duplicated resource.

8.1 Operation of GPSB

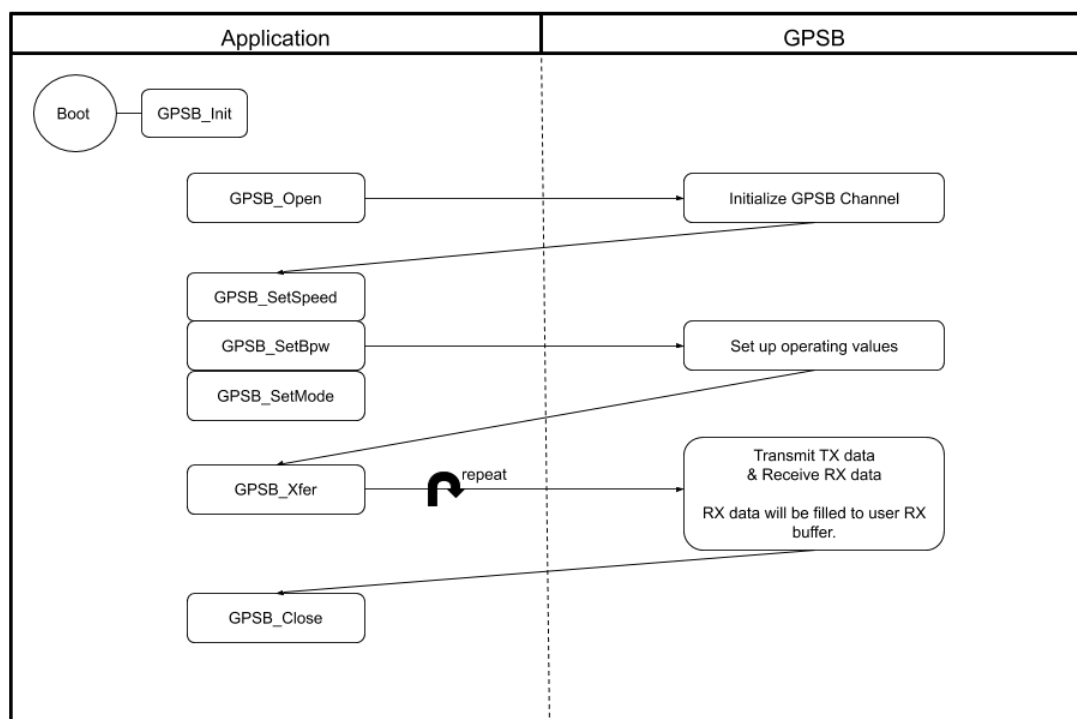


Figure 8.1 Diagram for GPSB

8.2 Example Code

Refer to test code in *gpsb_test.c* file.

```
static uint32    g_tx_dma;
static uint32    g_rx_dma;
static uint8     g_tx[GPSB_TEST_BUF_SIZE] = {0,};
static uint8     g_rx[GPSB_TEST_BUF_SIZE] = {0,};

{
    ucCh          = 0UL;
    dma_buf_size  = DMA_BUF_SIZE;
    gpio_sclk     = GPIO_GPB(25UL);
    gpio_cs       = GPIO_GPB(26UL);
    gpio_sdo      = GPIO_GPB(27UL);
    gpio_sdi      = GPIO_GPB(28UL);
    speed         = 20*1000*1000;
    ucBpw         = 16;
    cs_high       = 0;
    xfer_mode     = GPSB_XFER_MODE_WITHOUT_INTERRUPT;

    // initialize channel
    GPSB_Open(ucCh, gpio_sdo, gpio_sdi, gpio_sclk, g_tx_dma, g_rx_dma,
              dma_buf_size, NULL, NULL_PTR);

    // configuration
    GPSB_SetSpeed(ucCh, speed);
    GPSB_SetBpw(ucCh, ucBpw);
    GPSB_SetMode(ucCh, 0);

    // Set TX data in buffer (g_tx)
    GPSB_CsActivate(ucCh, gpio_cs, cs_high);
    GPSB_Xfer(ucCh, g_tx, g_rx, data_size, xfer_mode);
    GPSB_CsDeactivate(ucCh, gpio_cs, cs_high);

    // Check RX data in buffer (g_rx)

    // Disable channel
    GPSB_Close(ucCh);
}
```

9 REFERENCES

- [1] TCC805x Full Specification
- [2] Contact Telechips for more details: sales@telechips.com

10 REVISION HISTORY

Rev. 1.01: 2021-01-22

- Updated
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